

234054 F

Question 01

a) at equilibrium

$$Q_d = Q_s$$

Demand

$$P = 800 - 10Q$$

Supply

$$P = 5Q + 200$$

$$\therefore 800 - 10Q = 5Q + 200$$

$$600 = 15Q$$

$$Q = \frac{600}{15}$$

$$Q = 40$$

$$\text{Substitute } Q = 40$$

$$P = 800 - 10(40) = 800 - 400 \\ = 400$$

Equilibrium quantity = 40 rides

Equilibrium Price = 400 LKR*

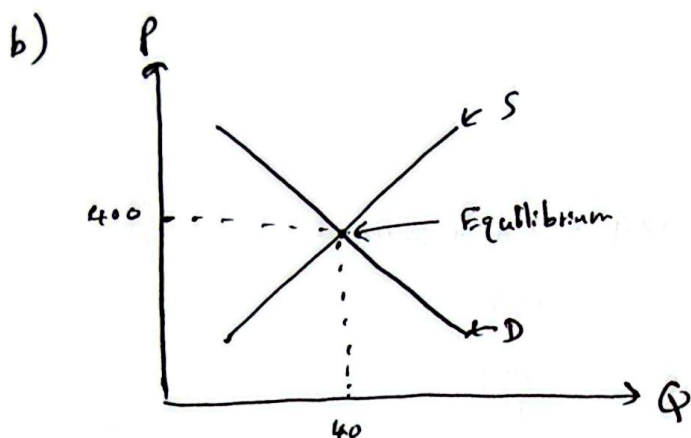
This equilibrium represents the point where:

- The no of rides passengers want = no of rides drivers supply
- There is no shortage or surplus.
- The market is efficient and stable.

In this ride hailing platform,

all passengers who want rides at 400 LKR get rides.

all drivers willing to work at 400 LKR are matched



c) Since the price cap equals equilibrium price,

- no shortage occurs
- no surplus occurs
- market continues to operate normally

So impact for passengers has no change.

impact for drivers has no change.

Therefore No economic distortion.

d) • Market price

• Demand increases $\rightarrow$ pushes Price $\uparrow$

• Supply decreases $\rightarrow$ pushes Price $\uparrow$

final effect $\Rightarrow$ Price increase

• No of rides

• Demand increases $\rightarrow$ increases quantity

• Supply decreases $\rightarrow$ decreases quantity

opposite effects ambiguous.

Quantity may increase or decrease depending on strength.

But usually $\rightarrow$ small change or uncertain.

